

Multimodal Large Model+robotic Arm Handling

Function Description

After the program runs, after waking up the voice module, input color block handling commands by voice. The large model will plan the actions to complete the color block handling task, then the program will control the robotic arm to handle the color blocks in order.

Startup

Open the terminal and enter the following command:

```
ros2 launch largemodel largemodel_control.launch.py
```

After waking up the module, input command by voice:

```
place the green block on the desktop above the red block
```

The available placement directions are: front, back, left, right, and top - these five directions.

```
root@yahboom:~/# ros2 run text_chat text_chat
user input: Place the green block on the desktop above the red block
okay@, let me think for a moment... \[INFO] [1775123877.111506338] [text_chat_node]: Decision making AI planning:1. Call 'seewhat()' to observe the environment and check the positions of t
he green block and the red block.
2. Call 'grasp_obj(x1, y1, x2, y2)' to grasp the green block, where '(x1, y1, x2, y2)' are the outer frame coordinates of the green block.
3. Call 'change_pose(x1, y1, x2, y2, "green", "red", 5, tar_size)' to place the green block on top of the red block, where '(x1, y1, x2, y2)' are the outer frame coordinates of the red block
, 'tar_size' indicates the height of the red block.
[INFO] [1775123888.813478479] [text_chat_node]: "action": ['seewhat()'], "response": "Okay, I'm going to place the green block on top of the red block. Let me first check where they are!
[INFO] [1775123889.251847512] [text_chat_node]: "action": ['grasp_obj(350, 220, 510, 380)'], "response": "I see the red block and the green block now. Let me grab the green one first!
[INFO] [1775123921.838441017] [text_chat_node]: "action": ["change_pose(30, 200, 170, 340, 'green', 'red', 5, 5)"], "response": "Got the green block! Now I'll place it right on top of the red
block.
[INFO] [1775123944.254675769] [text_chat_node]: "action": ['finishtask()'], "response": "Done! The green block is now perfectly placed on top of the red block. Everything's in order!
█
```

Task Planning

1. Call `seewhat()` to observe the environment and check the positions of the green block and the red block.
2. Call `grasp_obj(x1, y1, x2, y2)` to grasp the green block, where `(x1, y1, x2, y2)` are the outer frame coordinates of the green block.
3. Call `change_pose(x1, y1, x2, y2, "green", "red", 5, tar_size)` to place the green block on top of the red block, where `(x1, y1, x2, y2)` are the outer frame coordinates of the red block, `tar_size` indicates the height of the red block.

Core Code Analysis

Please refer to the **Core Code Analysis** content in the tutorial **【AI Model-Text Version】 - 【Multimodal Large Model + robotic Arm Handling】**. The voice version and text version have the same action functions; only the input method for task commands is different.

Advanced Play

We can have the robotic arm complete a more complex task: remember the current position of the color block, then place the color block on another color block, then put this block back to its original position. After waking up the module, input by voice:

Remember the current position of the red square, then place the red square on top of the yellow square, and finally return the red square to its original position.

Task Planning

1. Call `seewhat()` to observe the environment;
2. Call the `compute_pose(x1, y1, x2, y2)` function to record the current position of the yellow block;
3. Call `grasp_obj(x1, y1, x2, y2)` to grasp the yellow block, where x_1, y_1, x_2, y_2 are the outer frame coordinates of the yellow block;
4. Call `seewhat()` to observe the environment and check the position of the red block;
5. Call the `change_pose(x1, y1, x2, y2, "yellow", "red", side, tar_size)` function to place the yellow block on top of the red block, where x_1, y_1, x_2, y_2 are the outer frame coordinates of the red block, `side` indicates the placement position, and `tar_size` indicates the height of the red block;
6. Call `grasp_from_down_list("yellow")` to find the current position of the yellow block in the placement list;
7. Call the `return_to_orin("color")` function, where "color" is the color of the target block to be reset, valid values are "red", "blue", "green", or "yellow".

Core Code Analysis

Please refer to the **Core Code Analysis** content in the tutorial **【AI Model-Text Version】 - 【Multimodal Large Model + robotic Arm Handling】**. The voice version and text version have the same action functions; only the input method for task commands is different.